

Preeti Purnimaa Kannan

kannan.pr@northeastern.edu | (857) 415-8621 | Boston, MA
linkedin.com/in/preeti-purnimaa | github.com/preeti-purnimaa

Education

Northeastern University (GPA: 3.91)

Master of Science - Data Analytics Engineering

Sept 2024 - Aug 2026

Boston, MA

Anna University, Sri Venkateswara College of Engineering

Bachelor of Engineering - Computer Science and Engineering

Jun 2020 – Jun 2024

Chennai, India

Experience

Data Science Intern

MyEdMaster

Jan 2026 – Jun 2026

Remote, USA

- Reviewed 400+ peer-reviewed studies across 8 diet sub-domains for a proof-of-concept health recommendation engine model
- Designed Tableau dashboard tracking diet evidence quality across research domains for medical and executive stakeholders
- Engineered synthetic dataset in Python modeling diet adherence, macros, and health outcomes from literature effect sizes
- Built and validated Gradient Boosting model in scikit-learn predicting diet sub-score, supported SageMaker stacked deployment

Data Engineer Intern

Metaplore Solutions Pvt. Ltd.

Aug 2024 – Jan 2025

Chennai, India

- Built Python and SQL ETL pipelines standardizing transaction data across 3 enterprise systems, processing 400K daily records
- Performed data profiling and anomaly detection across 3 client source systems, improving overall data quality by 18%
- Designed 5 Power BI dashboards tracking transaction delay and exception volume KRIs, cutting manual reporting effort by 15%
- Optimized SQL transformation queries and restructured 2 pipeline tables, improving compliance reporting reliability further

Data Analyst Intern

CanBrs Therapeutics

May 2023 – Jul 2023

Chennai, India

- Consolidated raw plate-reader and HPLC exports from Azure Blob Storage into a validated Azure SQL database using Python
- Built validation logic in Python to catch missing wells, duplicate sample IDs, and out-of-range values before loading to SQL
- Modeled dose-response curves in Python to compute IC50 with 95% CI across 20+ assay batches, quantifying variability
- Delivered a Power BI dashboard with compound, batch, and timepoint drill-downs, surfacing 3 out-of-spec batches early

Data Scientist Intern

Exposys Data Labs

Nov 2021 – Mar 2022

Bangalore, India

- Built a multi-source ETL pipeline in Python and SQL, loading 110,000+ transaction records into a data warehouse
- Applied K-means clustering on RFM features to segment 9,600 customers into 5 behavior-driven groups for targeting
- Designed and ran a controlled A/B test on at-risk customers, applying hypothesis-driven analysis to guide outreach strategy
- Built an interactive Power BI dashboard with drill-downs by region, age, and segment to guide demographic marketing decisions

Projects

FortuneRAG: Serverless RAG Chatbot for Financial Analysis | AWS Bedrock, Claude-Sonnet, Lambda, Step Functions Aug 2026

- Architected a serverless RAG pipeline on AWS Bedrock, Claude-Sonnet, Lambda, and API Gateway for real-time financial Q&A
- Orchestrated automated ETL with AWS Step Functions and Glue, building a Bronze/Silver/Gold data pipeline in Amazon S3
- Engineered 7 Python Lambda functions and 3 REST API Gateway endpoints, fully integrated with a Streamlit web front-end
- Implemented semantic retrieval using vector embeddings on chunked live SEC filings for grounded, citation-backed LLM answers

Cross-Channel Marketing Performance & ROAS Analytics Platform | Python, SQL, Snowflake, Power BI Jun 2026

- Analyzed 1,800-row multi-platform ad dataset spanning Google, Meta, and TikTok Ads across 7 countries in Python
- Built SQL validation layer auditing metric integrity and duplicate campaign records, achieving 99.3% data accuracy
- Modeled ROAS and CAC by platform in SQL, uncovering TikTok's 7.64x ROAS versus Google's 3.48x on \$6.3M spend
- Delivered Power BI dashboard benchmarking \$11M ad spend across platforms, guiding data-driven budget reallocation

End-to-End DWBI Data Integration Project | Snowflake, Python, SQL, Power BI

Dec 2025

- Designed a full star schema in Snowflake with 5 dimension tables and a fact table across 100 store locations
- Automated generation of 6,000+ realistic test records across dimensions using Python, including customers, products, and stores
- Wrote scenario-based SQL queries in Snowflake to transform and validate data across all fact and dimension tables for accuracy
- Built a Power BI dashboard connected to Snowflake, surfacing revenue, regional performance, and customer segmentation

ELT Orders Data Pipeline with dbt, Snowflake, and Airflow | *Python, SQL, dbt, Snowflake, Airflow, Docker* **Aug 2025**

- Architected end-to-end ELT pipeline with dbt, Snowflake, and Airflow, orchestrating 5 models across 3 layers
- Automated daily orchestration via Airflow DAG using Astronomer Cosmos, auto-generating tasks from all 5 dbt models
- Modeled TPCB order data in dbt, joining staging sources into intermediate and fact tables to compute order sales metrics
- Enforced data quality with 6 dbt tests spanning generic and singular checks, validating uniqueness, nulls, and dates

Amazon Prime Movies & TV Shows Analytics Dashboard | *Power BI, DAX, Power Query, Excel* **Jun 2025**

- Built an end-to-end Power BI dashboard by transforming and modeling 9K+ Amazon Prime Movies & TV Shows records, creating an analytics-ready dataset using Power Query and DAX
- Implemented interactive dashboards with KPIs, slicers, and drill-through visualizations to analyze content distribution by genre, rating, release year, country, and type, improving business reporting and user exploration

Customer Behavior Analytics Dashboard | *Power BI, DAX, Power Query, Excel* **May 2026**

- Built an end-to-end customer analytics dashboard by cleaning, transforming, and modeling transactional data using Power Query and DAX to support interactive business reporting
- Implemented KPI metrics, cross-filtering, and interactive dashboards to analyze customer segmentation, purchasing behavior, revenue, sales performance, and subscription trends, enabling data-driven decision-making

Live Weather and Air Quality Analytics Platform | *Power BI, AWS S3, Glue, Athena, EC2, Python, SQL* **Dec 2025**

- Built an end-to-end AWS analytics platform ingesting hourly weather and air quality API data for 8 global cities, transforming raw JSON into curated Parquet datasets across 4 S3 data lake layers
- Created Glue catalog tables, 6 Athena reporting views, and a 5-page Power BI dashboard with AQI trends, PM2.5 monitoring, city comparison, pipeline health, and 7+ data-quality checks

EEG Seizure Detection Using Deep Learning | *Python, TensorFlow, Keras, CHB-MIT, Bonn EEG* **Jan 2025**

- Preprocessed 1M+ EEG signals from CHB-MIT and Bonn datasets to extract temporal and spectral features for seizure analysis
- Developed CNN-RNN hybrid model in Python/TensorFlow, achieving strong accuracy across cross-validated seizure detection
- Designed experimental pipeline with cross-validation and error analysis to compare models and improve generalisation
- Delivered prototype enabling early seizure alerts, supporting neurologist decision-making and real-time monitoring

California Wildfires Prediction & Analysis | *Python, Pandas, PostgreSQL, Tableau, ML* **Jun 2025**

- Built logistic regression model predicting California wildfires with 71% accuracy using Python and Pandas on 1,636+ fires
- Analyzed weather-fire correlations using PostgreSQL, identifying peak wildfire months through statistical analysis
- Engineered ETL pipeline processing 7 years of wildfire data, integrating weather APIs for feature enrichment
- Designed Tableau dashboards visualizing 4,000+ acres burned across counties, driving data-driven wildfire insights

Geospatial Data Analysis and Interactive Visualization | *Python, GeoPandas, Plotly, Scikit-learn, AWS S3* **Jan 2025**

- Analyzed 51K+ high-severity crash incidents in Tempe using GeoPandas to map spatial hotspots and identify risk corridors
- Applied DBSCAN clustering and temporal trend analysis to uncover patterns linked to weather, time, and location factors
- Built interactive Plotly dashboards with animated heatmaps, treemaps, and 3D hexbin maps for dynamic crash visualization
- Deployed AWS S3-hosted geospatial dashboards, enabling policymakers to make data-driven, proactive transportation decisions

Google Play Store & App Store Analysis Project | *Python, Pandas, Matplotlib, Tableau* **Aug 2025**

- Analyzed Google Play & App Store datasets using Python to compare ratings and reviews for launch-platform selection.
- Identified high-growth categories by evaluating ratings, engagement trends, and install patterns to guide app strategy.
- Assessed pricing models by comparing free vs paid adoption, revenue signals, and sentiment to optimize monetization choices.
- Built Tableau dashboards visualizing review-install correlations to provide clear platform-launch recommendations.

Walmart Sales Forecasting and Demand Analytics | *Python, EDA, Greykite, Neural Prophet* **Sep 2025**

- Performed EDA and preprocessing on Walmart sales data using pandas and seaborn to identify trends, seasonality, and outliers
- Engineered temporal features and applied Greykite and Neural Prophet models to forecast weekly store sales across 45 regions
- Assessed Silverkite and Neural Prophet models using RMSE and MAPE metrics, improving forecasting accuracy and reliability
- Built an ML pipeline in Python for data loading, training, and Flask deployment, ensuring reusable and scalable forecasting

Customer Segmentation and Behavioral Analytics | *Python, Matplotlib, Seaborn, RFM Framework* **Feb 2025**

- Performed RFM-based segmentation on eCommerce data, identifying high-value and dormant customers for retention.
- Applied K-Means clustering on standardized RFM scores, uncovering 3 customer segments that improved targeting by 25%.
- Visualized trends using bar charts, scatter plots, and heatmaps to reveal purchasing and seasonal behavior patterns.
- Delivered marketing recommendations by profiling clusters, enabling personalized engagement and higher ROI.

Airport Performance Optimization | *MySQL, MongoDB, Python, UML, EER Modeling* **Dec 2024**

- Analyzed airport operations data to identify inefficiencies and optimize real-time KPI tracking across multiple terminals, while designing scalable SQL/NoSQL database schemas using UML and EER modeling and automating Python ETL workflows to streamline performance reporting.
- Developed interactive Power BI dashboards to visualize flight delays, passenger throughput, and on-time performance metrics, improving data accuracy, reducing reporting delays, and enabling faster decision-making for airport management.

Intelligent Financial Reporting Agent | *GPT-4 (RAG), LangChain, Vanna AI, Microsoft Fabric* **Oct 2025**

- Architected a unified financial data warehouse on Microsoft Fabric OneLake to centralize diverse multi-source wealth data
- Implemented RAG-based GPT-4 pipelines to enable contextual, conversational financial analytics for portfolio performance
- Integrated Vanna AI text-to-SQL for natural-language querying, reducing manual financial reporting and analysis time by 60%
- Delivered an interactive insights dashboard offering real-time trends and comprehensive data-driven investment intelligence

Multimodal RAG-Based Food Recommendation System | *AWS Bedrock, FAISS, Claude-Sonnet* **Oct 2025**

- Developed a multimodal RAG framework on AWS Bedrock to combine text and image data for food recommendations
- Implemented FAISS with Titan Embeddings to encode and retrieve vectors, achieving real-time context-aware menu suggestions
- Integrated Claude-Sonnet model for image summarization and semantic understanding across cross-modal user queries
- Deployed an interactive Streamlit chatbot linking S3 storage and Bedrock to deliver adaptive AI-driven recommendations

Multi-Class Celebrity Detection and Classification System | *Python, YOLOv8, VGG16, ResNet50* **Aug 2025**

- Built CNN-based celebrity recognition system using YOLOv8 on CelebA dataset, achieving 98.3% mAP for detection
- Engineered balanced dataset of 7,050 augmented images across 47 classes to eliminate data imbalance issues
- Implemented transfer learning with VGG16/ResNet50, improving classification accuracy from 28.6% to 60%
- Generated synthetic multi-celebrity images via augmentation, enabling 95%+ confidence object detection

Fashion Product Image Generation Using Diffusion Models | *Stable Diffusion, Kandinsky, CLIP, FID, IS* **Dec 2025**

- Built an end-to-end text-to-image pipeline by cleaning 4k+ fashion samples for Stable Diffusion model training and testing.
- Performed 54+ tuning runs adjusting guidance, steps & schedulers to enhance prompt-image alignment and output quality.
- Applied CLIP, FID & Inception Score to evaluate images and identified Euler with guidance=9, steps=50 as optimal config.
- Benchmarked Stable Diffusion v1.5 vs Kandinsky 2.2 using CLIP scores & statistical tests to compare accuracy and stability.

AI/ML Mental Health Chatbot (SAAC) | *Python, NLP, Keras, TensorFlow, Web Speech API* **Apr 2024**

- Engineered an NLP-based mental health chatbot (SAAC) using Keras and TensorFlow, offering 24/7 personalized support
- Crafted a user-friendly interface with text and voice input via Web Speech API, enhancing accessibility
- Achieved 92% accuracy in intent classification, ensuring precise user interactions and effective mental health support
- Enabled chat history tracking, improving follow-up care and continuous user engagement

Real-Time 3D Human Pose Estimation for Ergonomics | *Python, YOLOv8, MediaPipe, Deep Learning* **Jan 2024**

- Built 3D posture estimation system using YOLOv8x-pose & MediaPipe, achieving 91.2% mAP for ergonomic monitoring
- Processed 500K+ kinematic keypoints to classify posture deviations and detect poor ergonomic alignment across frames
- Integrated live on-screen posture alerts to notify users during prolonged misalignment, reducing ergonomic strain risk
- Enhanced workplace health compliance by enabling real-time correction for text-neck and musculoskeletal disorder prevention

Skills

Programming & Querying: Python (NumPy, Pandas, Scikit-learn, GeoPandas), SQL, NoSQL, PySpark, Spark SQL, C

AI Frameworks & Libraries: PyTorch, TensorFlow, Transformers, Diffusers, OpenCV, spaCy

LLM & NLP: GPT-4, LangChain, RAG Pipelines, Vanna AI, Web Speech API, Intent Classification, Prompt Engineering

Data Visualization: Tableau, Power BI, Looker, MS Excel (Advanced, VBA, Power Query), Canva, Datawrapper

Data Engineering: ETL/ELT, Airflow, dbt, Snowflake, Streamlit, FastAPI

Cloud & Infrastructure: AWS (S3, Redshift, Glue, Athena, Lambda, QuickSight), Docker, Databricks